

- 8 (a) Two ions S and T, each of negative charge $-q$, are held stationary at a distance of 2.0 cm from each other as shown in Fig. 8.1. P_1 is the midpoint between S and T.

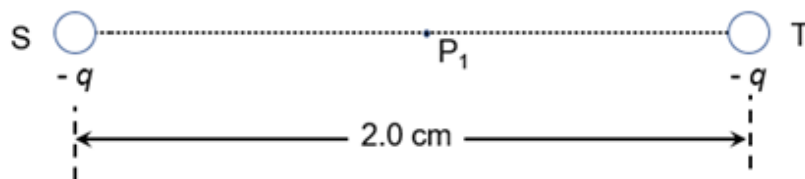


Fig. 8.1

State what it means by

- (i) the *electric field strength* at P_1 ,

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[1]

Solution:

It is the electric force per unit positive charge acting on a stationary test charge placed at P_1 due to the charges placed at S and T.

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- (ii) the *electric potential* at P_1 .

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[1]

Solution:

It is the work done per unit positive charge by an external agent in moving a point charge from infinity to P_1 .

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- (b) The value of q is known to be 1.6×10^{-19} C. A third ion R of positive charge $+q$ is introduced into the system at a distance of 4.0 cm from ions S and T as shown in Fig. 8.2. All three ions are held stationary.

P_2 is the midpoint between R and T.

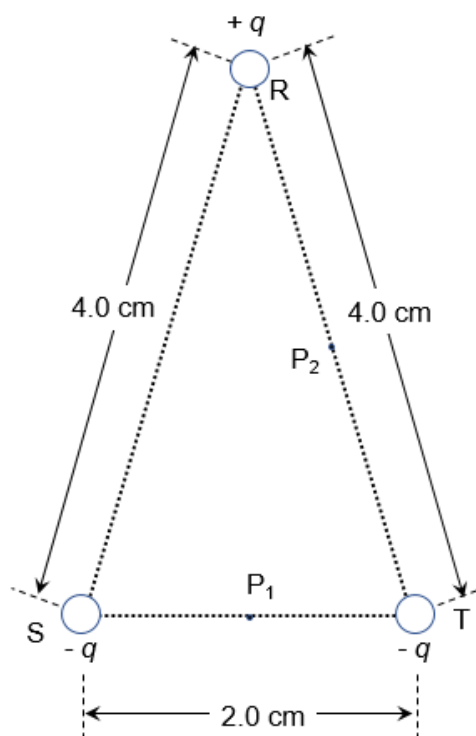


Fig. 8.2

- (i) Calculate the amount of work done required to assemble R.

work done = J [3]

Solution:

work done by external agent

$$= q_R V_{R,\text{net}}$$

$$= q_R (V_{RT} + V_{RS})$$

$$= q_R \left(\frac{q_S}{4\pi\epsilon_0 r_{RS}} + \frac{q_T}{4\pi\epsilon_0 r_{RT}} \right)$$

$$= \frac{q_R}{4\pi\epsilon_0 r_{RT}} (q_S + q_T) \quad \text{since } r_{RS} = r_{RT}$$

$$= \frac{+q}{4\pi\epsilon_0 r_{RT}} ((-q) + (-q))$$

$$= \frac{(1.6 \times 10^{-19})^2}{4\pi(8.85 \times 10^{-12})(4.0 \times 10^{-2})} (-2)$$

$$= -1.2 \times 10^{-26} \text{ J}$$

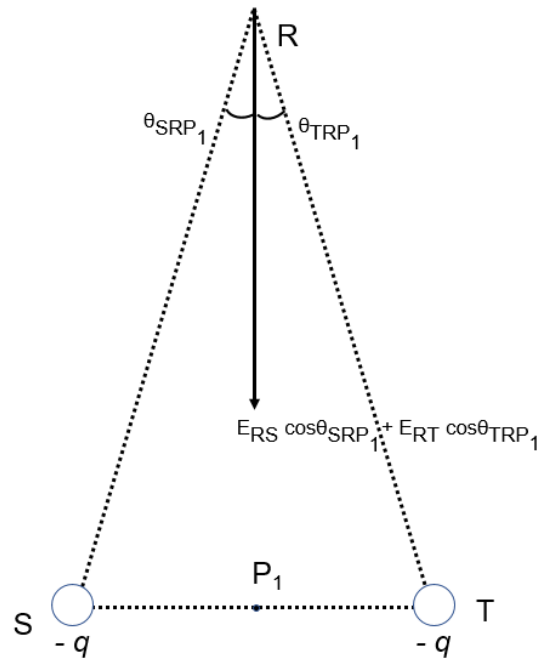
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- (ii) Without calculations, explain whether the work done required is more or less if R was placed at P_2 instead.

				
				[2]
			Solution: The resultant electric potential due to S and T at point P ₂ is more negative than at R's original position. Therefore an external agent needs to do more work in moving R from infinity to P ₂ .	1 1
		(iii)	With reference to the direction of the electric field strength along RP ₁ , describe the variation of electric potential along the line joining ion R to the point P ₁ .	
				
				[2]
			Solution: The resultant field strength is directed downwards everywhere along the line joining R and P ₁ and potential decreases in the direction of the field strength. (E= -dV/dr) The resultant potential along the line joining R and P ₁ decreases.	1 1
(c)	(i)		Show that the magnitude of the electric field strength that R experiences for the ion assembly in Fig. 8.2 is given by $1.74 \times 10^{-6} \text{ NC}^{-1}$.	
			Solution: $E_{RT} = \frac{q_T}{4\pi\epsilon_0 r_{RT}^2}$ $E_{RS} = \frac{q_S}{4\pi\epsilon_0 r_{RS}^2}$ $E = E_{RS} \cos\theta_{SRP_1} + E_{RT} \cos\theta_{TRP_1}$ $= \frac{q_S}{4\pi\epsilon_0 r_{RS}^2} \cos\theta_{SRP_1} + \frac{q_T}{4\pi\epsilon_0 r_{RT}^2} \cos\theta_{TRP_1}$	1 1



Since RST is an isosceles triangle,

$$r_{SR} = r_{RT} \text{ and } \cos \theta_{SRP_1} = \cos \theta_{TRP_1}$$

$$\begin{aligned} \cos \theta_{TRP_1} &= \frac{r_{RP_1}}{r_{RT}} \\ &= \frac{\sqrt{r_{RT}^2 - r_{P_1T}^2}}{r_{RT}} \\ &= \frac{\sqrt{4^2 - 1^2}}{4} \\ &= \frac{\sqrt{15}}{4} \end{aligned}$$

$$\begin{aligned} E &= \frac{q_S}{4\pi\epsilon_0 r_{RS}^2} \cos \theta_{SRP_1} + \frac{q_T}{4\pi\epsilon_0 r_{RT}^2} \cos \theta_{TRP_1} \\ &= 2 \frac{q}{4\pi\epsilon_0 r_{RT}^2} \cos \theta_{TRP_1} \\ &= 2 \frac{q}{4\pi\epsilon_0 (4.0 \times 10^{-2})^2} \left(\frac{\sqrt{15}}{4} \right) \\ &= 303 \frac{q}{\pi\epsilon_0} \\ &= 303 \frac{1.6 \times 10^{-19}}{\pi(8.85 \times 10^{-12})} \\ &= 1.74 \times 10^{-6} \text{ NC}^{-1} \end{aligned}$$

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[4]

(ii) R is subsequently released.

			Therefore R will oscillate to and fro about the line joining S and T. (1 mark) R will slow down because it experiences a force upwards. OR When R is released, it will accelerate towards P_1. Since all charges radiate when they accelerate, R will emit photons during its motion. Due to the conservation of momenta of R and the emitted photon, R will deviate from a straight path during photon emission.	1 (1) (1)
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